Team 06: Intellectual Conformity Under Autocracy

Research Question and Analysis Plan

AutoKnow Agent Orchestra — Team 06

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Research Question

Does autocracy reduce the intellectual diversity of SSH research, as measured by the semantic similarity of article abstracts within country-field-year clusters?

Rationale

The overarching question asks not only what topics are studied but what *direction* research takes and how knowledge *progresses*. A key dimension of scientific progress is intellectual diversity — the breadth of questions, frameworks, and arguments that scholars explore within a field. If autocracy constrains this diversity, it represents a qualitatively different form of impact than simply suppressing volume or shifting disciplinary composition. Semantic similarity of abstracts within country-field-year clusters provides a direct, content-based measure of whether the ideas being explored are more uniform under autocratic conditions.

Theoretical mechanism

Autocratic regimes constrain the intellectual diversity of SSH through three reinforcing channels. First, state censorship and editorial control narrow the range of acceptable arguments, questions, and conclusions — scholars cannot publish findings that contradict official narratives or threaten regime legitimacy. Second, anticipatory self-censorship leads scholars to cluster around "safe" framings and avoid intellectually risky or heterodox positions, even when direct repression is absent. Third, centralized research funding and agenda-setting (e.g., state-directed grant priorities, ideological guidelines for universities) channel scholars toward a narrower set of approved research questions within each field. Together, these mechanisms predict that abstracts from autocracies will be more semantically homogeneous — more similar to one another in content, framing, and argumentation — than abstracts from democracies in the same field and year.

Key parameters

Parameter	Value
Theory family	`intellectual-conformity`
Estimand	Within-country effect of `v2x_libdem` on semantic diversity
Unit of analysis	Country-field-year

Parameter	Value
Outcome variable	`semantic_diversity` (constructed from abstract embeddings)
Key independent variable	`v2x_libdem` (V-DEM Liberal Democracy Index, 0–1)

Analysis Plan

Method

Text embedding + OLS regression with two-way fixed effects. Article abstracts are embedded using the OpenAI `text-embedding-3-small` API (1536 dimensions). Within-cluster semantic diversity is computed at the country-field-year level. The main analysis regresses this diversity measure on `v2x_libdem`.

Data construction

Step 1: Embedding abstracts

- Embed all abstracts with non-NA text using the OpenAI Embeddings API (`text-embedding-3-small`).
- Estimated cost: approximately 2.5M abstracts at roughly 200 tokens each (500M tokens total, approximately $10).
- Batched processing in chunks of 1,000. Alternatively, a stratified random sample of 200,000 abstracts (stratified by country-regime-field-decade) can be used if full-corpus embedding is not desired.

Step 2: Computing semantic diversity

- Group articles by country × `subject_primary` × year.
- For each cell with ≥ 10 articles: compute the centroid embedding, then calculate the mean cosine distance from each article's embedding to the centroid.
- This avoids $O(n^2)$ pairwise computation. Mean distance-to-centroid captures dispersion of the embedding cloud.
- Secondary measure: standard deviation of pairwise cosine similarities (computed on a random subsample of pairs if $n > 50$).

Step 3: Sample restrictions

- Restrict to 1990–2023 (WOS coverage concern pre-1990).
- Restrict to country-field-year cells with ≥ 10 articles.
- Drop cells where all abstracts are NA.

Model specification

Main specification

$$\text{semantic\_diversity}_{c,f,t} = \beta \cdot \text{v2x\_libdem}_{c,t} + \alpha_c + \delta_f + \gamma_t + \mathbf{X}'_{c,f,t}\phi + \varepsilon_{c,f,t}$$

Component	Description
Outcome	`semantic_diversity` (mean cosine distance to centroid, continuous)
Key predictor	`v2x_libdem` (continuous, 0–1)
Fixed effects	Country (α_c) + field (δ_f) + year (γ_t)
Controls	log(n_articles) (cell size); mean author count per cell

Component	Description
Standard errors	Clustered at the country level
Expected sign of β	Positive (more democracy $\rightarrow$ greater semantic diversity)

Robustness specifications

1. Replace `v2x_libdem` with `regime_binary` (binary autocracy/democracy).
2. Use country-field fixed effects ($\alpha_{c,f}$) instead of separate country + field FE.
3. Restrict to the five largest SSH fields (Economics, Education, Psychology–Clinical, Business, Political Science).
4. Alternative diversity measure: standard deviation of pairwise cosine similarities.
5. Restrict to transition countries only (countries that change `v2x_regime` category during the period).

Causal identification strategy

Variation exploited

Within-country (or within-country-field) over-time variation in `v2x_libdem`. Country FE absorb time-invariant confounders (language, academic traditions, institutional structure). Year FE absorb global trends (WOS expansion, methodological convergence, field-level norms). The cell-size control addresses the mechanical relationship between number of articles and measured diversity.

Confounders controlled

- **Country FE:** national academic culture, language, institutional structure, historical field composition.
- **Field FE (or country-field FE):** discipline-specific norms for intellectual diversity, field size.
- **Year FE:** global trends in SSH publishing, WOS coverage changes, methodological shifts.
- **Cell size:** mechanical effect of n on diversity estimation.

Identification threats

1. **Economic development.** GDP growth drives both democratization and university expansion. Larger, better-funded systems may produce more diverse research regardless of regime type. The cell-size control partially addresses this.
2. **Internationalization.** Democracies may have more internationally connected scholars whose diverse training increases heterogeneity. This is a potential mediator rather than a pure confounder, but complicates interpretation.
3. **Journal indexing selection.** If WOS indexes a narrower slice of autocracies' research (e.g., only English-language output), observed abstracts may come from a more homogeneous subset. The transition-country robustness check helps (within-country regime changes with stable indexing patterns).
4. **Embedding artifacts.** Semantic similarity may capture stylistic or linguistic features rather than substantive content. High-dimensional embeddings and cosine distance mitigate this, but the concern remains.

5. **Field heterogeneity.** Different fields have different baseline diversity levels. Field FE or country-field FE address this.

Expected output files

File	Description
`fig1_diversity_by_regime.png`	Distribution of semantic diversity by regime category
`fig2_diversity_trends.png`	Time trends in semantic diversity, autocracies vs. democracies
`tab1_main_regression.png`	Main regression table with robustness checks
`fig3_field_heterogeneity.png`	Field-specific coefficient plot for top 5 SSH fields